WORKED SOLUTION

STEP III 1996, Question 12

Probability and Statistics • Markov Chains

STEP III 1996, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.16, *Markov Chains Forwards and Backwards*: https://beyondstep.keithmcnulty.org/ch12/markov.html

The key idea

This is a Markov chain: the type of the next theorem depends only on the type of the current one. Part (i) is the law of total probability, conditioning on the type of the current theorem. Part (ii) asks for a distribution that the chain leaves unchanged (a *stationary* distribution), which is a small linear system. Part (iii) runs the chain backwards, and reversing a conditional probability is exactly what Bayes' theorem does.

Solution

Number the theorems in order. Let X be the type of a given theorem and Y the type of the theorem after it, so that

$$\mathrm{P}(X = i) = a_i, \qquad \mathrm{P}(Y = j \mid X = i) = p_{ij}, \qquad \mathrm{P}(Y = j) = b_j.$$

Part (i)

The events $\{X = 1\}$, $\{X = 2\}$, $\{X = 3\}$ are mutually exclusive, and one of them must happen, because every theorem has exactly one of the three types. So they partition the sample space, and the event $\{Y = j\}$ splits into three disjoint pieces:

$$\{Y = j\} = \{X = 1,\, Y = j\} \cup \{X = 2,\, Y = j\} \cup \{X = 3,\, Y = j\}.$$

Adding the probabilities of the pieces, and writing each one as $\mathrm{P}(X = i)\,\mathrm{P}(Y = j \mid X = i)$,

$$b_j = \sum_{i=1}^{3} \mathrm{P}(X = i)\,\mathrm{P}(Y = j \mid X = i) = a_1 p_{1j} + a_2 p_{2j} + a_3 p_{3j},$$

which is the law of total probability. The given observation is used here: $\mathrm{P}(Y = j \mid X = i)$ is p_{ij} whatever happened before the current theorem.

Part (ii)

We need $b_j = a_j$ for each j, together with $a_1 + a_2 + a_3 = 1$, since the a_i form a probability distribution. Reading down the columns of the array, the three equations are

$$\begin{aligned} a_1 &= 0.3a_1 + 0.2a_2 + 0.1a_3, \\ a_2 &= 0.3a_1 + 0.4a_2 + 0.3a_3, \\ a_3 &= 0.4a_1 + 0.4a_2 + 0.6a_3. \end{aligned}$$

> **Ask yourself** Three equations in three unknowns, plus $a_1 + a_2 + a_3 = 1$: isn't that one equation too many?

It isn't, because the three equations are not independent. Adding them gives $a_1 + a_2 + a_3 = a_1 + a_2 + a_3$, since each row of the array sums to 1. So any two of them imply the third, and the normalising condition is what pins down the solution.

The middle equation is the easy one, because a_1 and a_3 have the same coefficient there. Using $a_1 + a_3 = 1 - a_2$,

$$a_2 = 0.3(1 - a_2) + 0.4a_2 \quad \Longrightarrow \quad 0.9a_2 = 0.3 \quad \Longrightarrow \quad a_2 = \tfrac{1}{3}.$$

Then $a_1 + a_3 = \frac{2}{3}$. Multiplying the first equation by 10 gives $7a_1 = 2a_2 + a_3 = \frac{2}{3} + a_3$, and substituting $a_3 = \frac{2}{3} - a_1$,

$$7a_1 = \tfrac{4}{3} - a_1 \quad \Longrightarrow \quad a_1 = \tfrac{1}{6}, \qquad a_3 = \tfrac{2}{3} - \tfrac{1}{6} = \tfrac{1}{2}.$$

Check the third equation: $0.4 \cdot \frac{1}{6} + 0.4 \cdot \frac{1}{3} + 0.6 \cdot \frac{1}{2} = \frac{1}{15} + \frac{2}{15} + \frac{3}{10} = \frac{1}{2} = a_3$. So

$$a_1 = \tfrac{1}{6}, \qquad a_2 = \tfrac{1}{3}, \qquad a_3 = \tfrac{1}{2}.$$

In the long run, then, half of Professor Ecks's theorems are false.

Part (iii)

Now take X to be the type of the preceding theorem and Y the type of the particular theorem, with X distributed as in part (ii). We want $q_{ij} = \mathrm{P}(X = i \mid Y = j)$. By Bayes' theorem,

$$q_{ij} = \frac{\mathrm{P}(X = i)\,\mathrm{P}(Y = j \mid X = i)}{\mathrm{P}(Y = j)} = \frac{a_i p_{ij}}{b_j} = \frac{a_i p_{ij}}{a_j},$$

where the last step uses $b_j = a_j$ from part (ii). This is why the question fixes these values of the a_i: then the particular theorem has the same distribution as the one before it.

Column by column:

- $j = 1$ ($a_1 = \frac{1}{6}$): $q_{11} = \dfrac{\frac{1}{6}\cdot 0.3}{\frac{1}{6}} = \frac{3}{10}$, $\quad q_{21} = \dfrac{\frac{1}{3}\cdot 0.2}{\frac{1}{6}} = \frac{2}{5}$, $\quad q_{31} = \dfrac{\frac{1}{2}\cdot 0.1}{\frac{1}{6}} = \frac{3}{10}$.
- $j = 2$ ($a_2 = \frac{1}{3}$): $q_{12} = \dfrac{\frac{1}{6}\cdot 0.3}{\frac{1}{3}} = \frac{3}{20}$, $\quad q_{22} = 0.4 = \frac{2}{5}$, $\quad q_{32} = \dfrac{\frac{1}{2}\cdot 0.3}{\frac{1}{3}} = \frac{9}{20}$.
- $j = 3$ ($a_3 = \frac{1}{2}$): $q_{13} = \dfrac{\frac{1}{6}\cdot 0.4}{\frac{1}{2}} = \frac{2}{15}$, $\quad q_{23} = \dfrac{\frac{1}{3}\cdot 0.4}{\frac{1}{2}} = \frac{4}{15}$, $\quad q_{33} = 0.6 = \frac{3}{5}$.

Arranged like the original array (row i, column j):

$$(q_{ij}) = \begin{pmatrix} \frac{3}{10} & \frac{3}{20} & \frac{2}{15} \\ \frac{2}{5} & \frac{2}{5} & \frac{4}{15} \\ \frac{3}{10} & \frac{9}{20} & \frac{3}{5} \end{pmatrix} = \begin{pmatrix} 0.3 & 0.15 & 0.1\dot{3} \\ 0.4 & 0.4 & 0.2\dot{6} \\ 0.3 & 0.45 & 0.6 \end{pmatrix}.$$

Here it is the *columns* that sum to 1: for a fixed type j of the particular theorem, the preceding theorem must have been of some type.

Checks

- Each column of (q_{ij}) sums to 1; for example $\frac{2}{15} + \frac{4}{15} + \frac{3}{5} = 1$.
- The diagonal entries satisfy $q_{ii} = p_{ii}$, as the formula $q_{ii} = a_i p_{ii} / a_i$ predicts.
- A simulation of 10^6 steps of the chain gave long-run frequencies 0.167, 0.333, 0.501 and backward proportions within 0.001 of every q_{ij} above.

Common slips

- Multiplying the array by a column vector of the a_i (using rows instead of columns). The equation is $b_j = \sum_i a_i p_{ij}$, so the sum runs down column j.
- Forgetting $a_1 + a_2 + a_3 = 1$, and then being puzzled that the three balance equations do not determine the a_i.
- Answering part (iii) with p_{ji}. Going backwards in time is not the same as reading the array the other way round; this chain is not reversible, since $q_{12} = 0.15$ but $p_{12} = 0.3$.

Beyond STEP

The vector (a_1, a_2, a_3) is the stationary distribution of the chain: a left eigenvector of the transition matrix with eigenvalue 1. Because every entry of the matrix is positive, the distribution of the type of the nth theorem converges to it whatever type the first theorem had. The array (q_{ij}) describes the *time-reversed* chain, and chains whose reversal looks the same as the original (with $a_i p_{ij} = a_j p_{ji}$, the detailed balance condition) are central to random walks and to Monte Carlo methods. Grimmett and Welsh, *Probability: An Introduction*, has an accessible chapter on Markov chains.

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